Multimedia Signal Processing - 2nd Module

July, 10 2018

1. **Question (8 pts)**
   Consider a discrete memoryless source X that emits symbols from the alphabet $\{a, b, c\}$ according to the following p.m.f.:

symbol	Pr.
a	0.5
b	0.4
c	0.1

   (a) Determine the entropy of the source and lower/upper bounds on the average code length of a Shannon code that encodes $N = 1$ symbols at the time.

   (b) Build a Huffman code for this source ($N = 1$) and compute its average code length. Compare such length with the bounds above.

   (c) Consider a code that encodes $N = 2$ symbols at the time. Revise the lower/upper bounds above, build a Huffman code and compare its length with the revised bounds.

   (d) Consider the following sequences of symbols: *bccb*. Compute the number of bits per symbol using the Huffman codes above ($N = 1$ and $N = 2$) and compare it with the bounds. Comment the result.

2. **Question (5 pts)**
   (a) Illustrate the block diagram of a hybrid DCT/DPCM video encoder, commenting each block.

   (b) Explain a possible way of computing motion vectors, and how they are exploited in a video coding scheme.

3. **Question (5 pts)**
   (a) Introduce the concept of scalar quantization, providing the definition of scalar quantizer and showing the difference between a midtread and a midrise one.

   (b) Describe the compander/expander operation, highlighting how quantization can benefit from it.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**
   (a) Consider the quantization process of a source X emitting symbols from an infinite alphabet with probability distribution function $p(x)$.
   - [T] [F] A scalar uniform quantizer cannot be used if $p(x)$ is not a uniform distribution.
   - [T] [F] A scalar uniform quantizer is an optimal choice if $p(x)$ is a uniform distribution.
   - [T] [F] In a Lloyd-Max iteration, once reproduction levels have been updated, decision regions update does not depend on $p(x)$.
   - [T] [F] Lloyd-Max quantizer always provide better results than a uniform scalar quantizer.

   (b) Consider the problem of speech coding.
   - [T] [F] The spectral envelop of unvoiced signals is characterized by a set of formants.
   - [T] [F] When using a DPCM, the long-term predictor helps compensating the pitch structure of the voiced spectrum.
   - [T] [F] When using linear predictive vocoder exploiting the source-system model, the coefficients of the synthesis filter are directly quantized.

   (c) Consider JPEG image coding scheme.
   - [T] [F] KLT is used as preferred transform coding method.
   - [T] [F] Transform coding is applied to 8×8 blocks.
   - [T] [F] Luminance and chrominance components are always given the same importance.

5. **Question (Matlab - 10pts)**
   Consider the image file `lena.png`.
   (a) Load the 8 bit grayscale image `I` and display it on screen.
   (b) Consider the image as a single sequence of pixels. Display the probability mass function (pmf) of the pixels and compute the entropy. Is it compatible with the image being stored using 8 bit per pixel?
   (c) Consider the image as a series of distinct blocks of two adjacent pixels. Display the 2D probability mass function (pmf) and compute the entropy. Compare with the previously obtained result and comment.
   (d) Encode `I` using block-wise transform coding based on 8×8 DCT and draw rate-distortion curve. Specifically, consider:
   - scalar uniform quantization with $Q \in \{8, 12, 14, 16, 20\}$ (hint: use `round(.)` and ensure a reproduction level equal to zero).
   - PSNR as distortion measure.
   - Entropy as rate measure.

   (e) Rapidly comment the rate-distortion curve obtained (hint: what happens for $Q = 14$?).
   (f) Considering $Q = 14$, compute and display image compression residual (i.e., difference between original image and decoded one). Comment what you see.